pAI NeurIPS 2026 — Submission Gap Plan

Changes to Technical Report — Changes to Muon Paper

Abstract: May 4 AoE Full paper: May 6 AoE Today: April 23, 2026

Internal — Do Not Circulate

Framing. The NeurIPS submission is the **pAI Technical Report**. The Muon paper (v5, `v5_*.tex`) is the live-demonstration artefact produced *by* pAI. Both documents need changes, but for different reasons: the tech report needs its empty tables, missing theorems, and missing experiments filled; the Muon paper needs its own open mathematical items resolved and its framing sharpened.

Key finding. Muon v5 is far more complete than the tech report implied. Theorem A (all four parts), Theorem B (isotropic formula), and the proofs are fully drafted in the `.tex` files. The tech report needs to *extract* and *correctly present* this content — not regenerate it.

Budget. Total: \$6k. One full pAI run: \$4k. **Do not run a second full pipeline.** Estimated spend on all fixes below: \$305–\$1,520, leaving \$4,480–\$5,695 in reserve.

Item	Min	Max
Single-prompt baseline API calls (Fix T3)	\$5	\$20
Ablation partial runs, 6 runs at \$50–150 (Fix T5)	\$300	\$800
Optional: theory sub-run for anisotropic Theorem B (Fix M3)	\$0	\$200
Optional: discrete-time Campaign C4 (Fix M5)	\$0	\$300
Total	**\$305**	**\$1,520**

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Part I

Changes to the pAI Technical Report

Completion target. All P1 items must be done by April 30. P2 items by May 2. P3 items by May 3, before anonymisation and final compile. The submission is the tech report; every fix in this part changes `pAI_tech_report.tex` directly.

1 [T1] P1 Correct the Frobenius Norm Error in Theorem 1(ii)

Due: April 24 **Cost:** \$0 **Effort:** 30 min

What is wrong

The tech report states Theorem 1(ii) as: "the Frobenius norm is *conserved*, $\frac{d}{dt}\|W(t)\|_F^2 = 0$." This is factually wrong. The Muon v5 proof (`appendix_proofs-15.tex`) shows:

$$\frac{d}{dt}\|W(t)\|_F^2 = -2\eta \sum_i \alpha_i \sigma_i \;\leq\; 0.$$

The norm *contracts* monotonically; it is not conserved. This is also the error that the Lyapunov council episode corrected (the cycle-3 corrected theorem in the Muon proofs is named `GPT54_CYCLE3_corrected_theorem`).

Deliverables

1. In §5.3: replace the theorem statement with $\frac{d}{dt}\|W(t)\|_F^2 = -2\eta \sum_i \alpha_i \sigma_i \leq 0$.
2. Search and replace every occurrence of "Frobenius-conserving" or "norm conservation" in the tech report body.
3. In §5.4 (Lyapunov episode narrative): clarify that the error the council detected was precisely this conservation/contraction conflation, and that the corrected version appears as `GPT54_CYCLE3_corrected_theorem` in the Muon proofs appendix.

2 [T2] P1 Fill All Theorem Placeholders from Muon v5

Due: April 26 **Cost:** \$0 **Effort:** 3–4 hr

What is wrong

The tech report has `[GAP]` for Theorem 1(iii) (sign criterion), the entire Theorem 2 statement, and `[NUM]` for Proposition 3. All of these exist in full in the Muon v5 `.tex` files and need only to be extracted.

Deliverables

1. **Theorem 1(iii) sign criterion.** From `v5_theory-13.tex`, transcribe:

$$\frac{d\widetilde{R}_{\mathrm{pw}}}{dt} = \frac{2\eta}{\rho} \sum_{i=1}^{r} \alpha_i \big(S_i - C\tilde{\sigma}_i\big).$$

State that spectral flattening requires this quantity to be ≤ 0; that this is *not* universally true (counterexample: uniform α_i); and that it is empirically verified across all tested configurations.

2. **Theorem 2 (Critical Batch Size).** Transcribe both formulas from `v5_theory-13.tex`:

$$B_{\text{crit}} = \frac{1-\beta}{1+\beta} \cdot \frac{\text{tr}(D_P[\Sigma]D_P^\top)}{r} \quad \text{(general)}; \qquad B_{\text{crit}} = \frac{2\sigma^2 S(\mu)}{r} \quad \text{(isotropic, } \beta = 0\text{)}.$$

Include the "proved vs. heuristic" caveat from Muon v5 §3.4: the SDE approximation and the signal=noise crossing criterion are heuristic. The anisotropic extension is at *proposed* status. Report the three numerical validations from `v5_results-12.tex`: $S(\mu)$ exact for $r \in \{2, 4, 8, 16\}$; momentum correction 10.3× difference from naive scaling; B_{crit} formula confirmed across four spectrum types.

3. **Proposition 3.** Extract the exact NanoGPT-124M misalignment numbers from `v5_transformer-14.tex`. Fill `[NUM:`$\theta°$`]`. State CI type, sample size (layers × seeds), and statistical test. Add the two-condition disclaimer from Muon v5: requires (a) effective underdetermination via low activation rank and (b) local linearisation — both only approximately valid. Frame NanoGPT observations as "suggestive evidence, not theorem-level validation."

3 [T3] P1 Run Single-Prompt Baseline (Table 3)

Due: April 25 **Cost:** <$20 **Effort:** 2 hr

What is wrong

Table 3 is entirely `[GAP:?]`. This table is the primary counterfactual for the paper's central claim. Without it, there is no empirical evidence that pAI adds value over a single frontier-model call.

Deliverables

1. Write one prompt giving: (a) the Muon regression setup (as in Muon v5 §2), (b) the key prior papers, (c) an explicit instruction to produce a formally stated theorem with proof and empirical predictions. Save as `eval/baseline_prompt_v1.txt`.
2. Run against GPT-5.4, Claude Opus 4.6, Gemini 3.1 Pro, and o3 via direct API (temperature 0, single call each). Save raw responses to `eval/baseline_responses/{model}_v1.txt`.
3. Score each response on five binary criteria grounded in Muon v5 ground truth:
   - C1: Formally stated theorem produced?
   - C2: Theorem correct? Specifically: identifies polar regulariser R_{pw} (not nuclear norm) and states Frobenius *contraction* (not conservation)?
   - C3: Proof of $\dot{\sigma}_i = -\eta\alpha_i$ present and gap-free?
   - C4: Empirical prediction specific enough to verify (e.g., a B_{crit} formula)?
   - C5: Self-detection and correction of any error?
4. Fill Table 3. The pAI row is: C1=Yes, C2=Yes (council corrected), C3=Yes (full proof in appendix), C4=Yes (B_{crit} to machine precision), C5=Yes (Lyapunov episode).

5. Write the §6.1 interpretation paragraph connecting each failed criterion to a specific pAI mechanism: C2/C5 failures → adversarial cross-family council; C3 failures → claim-DAG lifecycle; C4 failures → dual-track experiment coupling.

4 [T4] P1 Fill Campaign Trace Table 2 and Figure 2

Due: April 25 **Cost:** $0 **Effort:** 2 hr

What is wrong

Every P/VS/VN/A cell in Table 2 is `[GAP:n]`. The data exists in `logs/` at `MSc_Internal@6fe0e8e`.

Deliverables

1. Write `scripts/extract_dag_counts.py`: parse DAG state snapshots at each milestone gate timestamp; output CSV with columns `steer_id, gate, P, VS, VN, A`.
2. Fill Table 2. Verify Steer 4 shows a step-change in P (five nodes cascade back to *proposed* upon nuclear-norm falsification). Verify final state matches Muon v5: Theorems A and B at *accepted*; anisotropic extension at *proposed*.
3. Add Figure 2 (two panels): (a) stacked bar chart of P/VS/VN/A counts across 10 steers; (b) cumulative API cost burn-down with milestone gate markers. Save as `figures/fig2_campaign_trace.pdf`.
4. Write Table 2 caption so the Steer 4 nuclear-norm cascade is self-contained in the caption alone.

5 [T5] P1 Ablation Study (6 Partial Runs)

Due: April 30 **Cost:** $300–$800 **Effort:** 2–3 days

What is wrong

The tech report claims ablation results exist in Appendix ??. That appendix does not exist. This is the primary empirical weakness; NeurIPS reviewers will raise it immediately. A full 12-run ablation is infeasible ($48k). Six partial runs (manuscript stage disabled, reduced model routing, $50–150 each) is the minimum credible ablation.

Ablation Design

Three configurations × two held-out problems = 6 runs.

Configurations:

- **A: Full system.** All four mechanisms active.
- **B: −Council.** Cross-family council replaced by single-model verifier (Claude Sonnet 4.6).
- **C: −Dual-track.** Theory and experiment run sequentially (theory first, then experiment post-hoc).

Held-out problems (with known ground truth):

- **P1: GD implicit bias** — minimum nuclear norm (Gunasekar et al. 2017). Direct foil to the Muon finding.

- **P2: Adam implicit bias** — ℓ^1/max-norm (Jiang et al. 2023 or equivalent). Tests generalisation beyond Muon.

Metrics per run (binary):

- M1: Correct theorem statement vs. known ground truth?
- M2: Self-correction of at least one non-trivial error without human intervention?
- M3: Empirical prediction verifiable within 5%?

Deliverables

1. Configure 6 runs: `ablation/configs/{full,no_council,no_dualtrack}_{p1,p2}.yaml`.
2. Score all runs on M1–M3; record in `ablation/results/scores.csv`.
3. Write **Appendix F**: description of held-out problems and ground truth; results table (6 rows × 3 metrics); one paragraph per configuration connecting the delta to Muon-campaign anecdotal evidence (council → Lyapunov episode; dual-track → nuclear-norm falsification at Steer 4).
4. Update §6.2: replace `[VERIFY: Appendix ??]` with a forward reference to Appendix F and a one-sentence headline finding.

Fallback: if not all 6 runs finish by April 30, report completed runs and mark the rest as "in progress, to be completed for camera-ready." Config A (full system, both problems) must complete first.

6 [T6] P1 Extract Lyapunov Council Artefacts (Appendix G)

Due: April 26 **Cost:** $0 **Effort:** 2 hr

What is wrong

Section 5.4 narrates the Lyapunov episode but all artefacts are tagged `[VERIFY: Reproduce from council_lyapunov_2025-11-14.json]`. The episode is the paper's strongest evidence for the council mechanism but reads as anecdote without the actual artefacts.

Deliverables

1. Extract the Python falsification script from `campaign_artefacts/council_lyapunov_2025-11-14.json`; verify it produces the positive $\frac{d}{dt}\|W\|_F^2$ value for the stated ($d = 4, r = 2, \sigma_1/\sigma_2 = 3.1$) configuration; format as `lstlisting` in the appendix.
2. Create the council vote table in Appendix G: Member / Verdict / Justification / Minority report preserved? Four rows: GPT-5.4 (REJECT), Claude Opus 4.6 (REJECT), Gemini 3.1 Pro (REVISE with proposed ≤ 0 correction), Synthesis (3:1 REJECT, Gemini minority report preserved).
3. Quote the Gemini minority report verbatim ($\leq$ 10 lines) to demonstrate minority-report preservation.
4. Add a four-node cascade diagram showing nodes v1–v4 of Theorem A propagated to *rejected* in a single DAG step, with node IDs from the log.
5. Update §5.4: replace all `[VERIFY]` tags with forward references to Appendix G and the `lstlisting`.

7 [T7] P1 Fill Cost and Efficiency Numbers (§6.3)

Due: April 25 **Cost:** $0 **Effort:** 1 hr

What is wrong

Section 6.3 has `[NUM: Report total API cost, wall-clock time, HPC compute, LLM call count from logs/cost_summary.json]`. These numbers exist in the logs.

Deliverables

1. Parse `logs/cost_summary.json`: extract total API cost ($), wall-clock time (hours), HPC compute (GPU-hours), LLM call count by model.
2. Fill §6.3 with all four numbers with units.
3. State the human-to-machine ratio: 10 steers $\times$ < 5 minutes =< 50 minutes human attention vs. X GPU-hours machine compute. Compare to the $\sim 10^3$ prompt estimate for an ad-hoc workflow.
4. Add a footnote to Table 1 (positioning table) stating the USD cost of the Muon campaign.

8 [T8] P1 Rebuild `.bib` and Resolve All ? Citations

Due: April 27 **Cost:** $0 **Effort:** 3 hr

What is wrong

The tech report has ∼40–60 unresolved `?` citation placeholders. A paper submitted with `?` citations is desk-rejected.

Deliverables

1. Run `grep -n "(?)" pAI_tech_report.tex` wc -l—. Record count and locations in `bib_audit.txt`.
2. Copy all entries already resolved in the Muon paper's `references-3.bib`: Muon optimiser, Moonlight scaling, McCandlish B_{crit}, Gunasekar nuclear-norm, Mathias polar Fréchet, matrix-sensing papers, NanoGPT.
3. Resolve remaining entries not in the Muon bib:
   - AI-for-research systems: AI Scientist v1/v2, AIDE, AlphaEvolve, AICo-Scientist, Aletheia, AxProverBase, FunSearch, AlphaProof/Geometry, LeanDojo/ReProver.
   - Multi-agent debate: Du et al., Liang et al., and a third.
   - CBMM Technical Memo 160 (pAI architecture preview).
   - The 42%/57% AI Scientist hallucination audit study.
   - The $\sim 10^3$ prompts per manuscript effort estimate.
4. Build merged `refs_merged.bib`. Compile and verify: `pdflatex; bibtex; pdflatex; pdflatex`. Zero undefined references.

9 [T9] P2 Add ATSR and Weight-Decay Results (§5.3)

Due: April 28 **Cost:** $0 **Effort:** 1 hr

What is wrong

The tech report does not present the ATSR phase-diagram result, which is one of the strongest empirical outputs of the Muon campaign. The figures and numbers already exist in `v5_results-12.tex`.

Deliverables

1. Add a paragraph in §5.3 summarising the ATSR phase diagram: coupled WD at $\lambda = 0.01$: ATSR = 7.54 (4.12× degradation); decoupled WD: ATSR = 2.28 (1.25×); baseline: 1.83. Phase boundary at $\lambda \approx 0.01$.
2. Insert Figure 3 into the tech report: reference `figures/fig_atsr_phase_diagram.pdf` from the Muon paper.
3. Add one sentence noting that this result was produced by the experiment track concurrently with the theory track, demonstrating dual-track bidirectional DAG state exchange.

10 [T10] P2 Implementation Maturity Table (Appendix E)

Due: April 28 **Cost:** $0 **Effort:** 2 hr

What is wrong

Section 4.4 has `[VERIFY: Confirm whether gates enforce hard schema or prompt-constructed contracts]`. This is the most common reviewer attack on systems papers: "your artefact contracts are aspirational." An honest table pre-empts it.

Deliverables

1. Build Table E.1 in Appendix E with columns: Gate name / Paper claim / Implemented? (Yes/Partial/No) / Schema enforcement type (Hard JSON / Prompt-constructed / None) / Evidence location (file path, commit hash). One row for each of the seven gates: `lit-gate`, `brainstorm-gate`, `track-decomp`, `duality-gate`, `writeup-gate`, `proofread-gate`, `review-gate`.
2. Be honest: if some gates are prompt-constructed, say so. Frame it as "X gates use hard JSON schema; Y gates use prompt-constructed contracts; strengthening the latter is future work."
3. Update §4.4: replace `[VERIFY]` with forward reference to Table E.1.

11 [T11] P2 External Expert Proof Verification

Due: Send by April 25; receive by May 2 **Cost:** $0

What is wrong

The tech report flags Open Item P10: expert verification not on author list required before submission.

Deliverables

1. Prepare a self-contained verification package by April 25: Theorem A (all four parts) with full proof from `appendix_proofs-15.tex`, Theorem B proof sketch, and the numerical sanity-check scripts.

2. Contact at least two external experts (matrix ODE analysis, Riemannian optimisation, or implicit bias). Request two-week turnaround.
3. Key questions: (a) Is the sign-criterion proof (steps 3–7) complete and gap-free? (b) Is the $r = 2$ edge case handled correctly? (c) Is the AR(1) Lyapunov equation in Theorem B correctly derived?
4. Record the outcome in the tech report acknowledgements. If an error is found, escalate to a targeted pAI theory sub-run ($50–$200).

12 [T12] P3 Anonymisation

Due: May 3 **Cost:** $0 **Effort:** 1 hr

Deliverables

1. Remove author names; replace with "Anonymous Authors."
2. Remove MIT, CBMM, Poggio Lab, Perseus Labs from all locations.
3. Replace `github.com/PoggioAI/PoggioAI_MSc` with an anonymised repo link (anonymous GitHub org or Zenodo DOI).
4. Anonymise or omit the CBMM Technical Memo 160 reference.
5. Remove `mabdel03@mit.edu` and all identifying emails.
6. Final check: `grep -i "poggio\`MIT$\|CBMM\|Abdelmoneum\|Beneventano\|Perseus"pAI_tech_report.tex|n$

13 [T13] P3 NeurIPS Checklist and Broader Impact

Due: May 3 **Cost:** $0 **Effort:** 1–2 hr

Deliverables

1. Complete the NeurIPS 2026 paper checklist (download from `https://neurips.cc/Conferences/2026`). Flag Theorem B anisotropic extension as open; flag Proposition D as conjectural; report compute cost from Fix T7.
2. Expand §7 (Broader Impact) to three paragraphs: (a) positive effects (accelerated ML theory); (b) risk of over-trusting automated proofs, with mitigation (the four named human decision gates); (c) open-source release and research-monoculture risk.
3. Attach completed checklist PDF as supplementary material.

14 [T14] P3 Abstract and Contribution List Rewrite

Due: May 3 **Cost:** $0 **Effort:** 1–2 hr

Do this last, after all other fixes are complete.

Deliverables

1. Rewrite the abstract to replace generic language with specific numbers now available:
   - Theorem A: "$\dot{\sigma}_i = -\eta\alpha_i$ and a spectral-flattening sign criterion (not nuclear norm)."

- Theorem B: "$B_{\text{crit}} = 2\sigma^2 S(\mu)/r$ verified to machine precision for $r \in \{2, 4, 8, 16\}$."
- Council: "full council artefacts (script, vote record, minority report) in Appendix G."
- Table 3: state the actual finding (e.g., "no single frontier-model call produced a correct theorem statement").
- Ablation: "six-run partial ablation (Appendix F) shows removing the council eliminates autonomous error correction across both held-out problems."

2. Update the contribution list: add evidence pointers (table/figure/appendix numbers) to every contribution.

3. Ensure abstract is ≤ 250 words: `texcount -1 pAI_tech_report.tex`.

Part II

Changes to the Muon Paper

Context. The Muon paper is the demonstration artefact produced by pAI. It is not the NeurIPS submission, but it is cited by the tech report and may be released as supplementary material. Its own mathematical open items must be resolved so that the tech report can accurately report what pAI produced.

Current status from reading v5 `.tex` files. Theorem A (all four parts) and Theorem B (isotropic formula) are fully drafted with complete proofs. The anisotropic Theorem B extension is at *proposed* status. Proposition D is correctly flagged as conjectural. The experimental validation is complete and figures exist. The remaining changes are targeted, not a full rewrite.

15 [M1] P1 Verify Theorem A(ii) Wording Throughout

Due: April 24 **Cost:** $0 **Effort:** 20 min

What to do

The Muon v5 proof is correct (Frobenius contraction, ≤ 0). Confirm no stale "conservation" language remains anywhere in the `v5_*.tex` files from earlier draft cycles.

1. Run: `grep -rn "conserved\`conservation" v5$_*$.*tex*|

1. If any hit refers to the Frobenius norm, correct it to "monotonically nonincreasing."

2. No other changes needed to Theorem A.

16 [M2] P1 Resolve All ? Citations in Muon Paper

Due: April 27 **Cost:** $0 **Effort:** 2 hr

What to do

1. Run: `grep -rn "(?)" v5_*.tex` wc -l—. Record locations.

2. Resolve all unresolved entries in `references-3.bib`. The merged bib from Fix T8 should supply most of them; the Muon paper additionally needs concurrent Muon implicit-bias citations (see Fix M4 below).

3. Compile Muon paper and verify: zero `?` in the PDF.

17 [M3] P2 Theorem B Anisotropic Extension

Due: April 29 **Cost:** $0–$200 **Effort:** 2–4 hr

What is wrong

`L_anisotropic_sigma_sensitivity` — the closed-form expansion of $\mathrm{tr}(D_P[\Sigma]D_P^\top)$ for non-isotropic Σ — is at *proposed* status in the claim-DAG. The proof in `appendix_proofs-15.tex`

ends with the note: "the anisotropic extension requires further work." This is an honest limitation but it must be stated explicitly in the main text, not just the appendix.

Deliverables

1. **Option A (preferred):** Run a targeted pAI theory-track sub-run ($50–$200) to push `L_anisotropic_sigma_sensitivity` to *verified_numeric*. Update Theorem B accordingly.
2. **Option B (if no sub-run):** Add a one-paragraph discussion in Appendix B: scope the anisotropic case as open; give a concrete proof strategy (apply the Mathias formula for non-symmetric Σ; the missing step is bounding $\mathrm{tr}(D_P[\Sigma]D_P^\top)$ in terms of eigenvalues of Σ).
3. In both options: ensure the "proved vs. heuristic" paragraph in §3.4 is present in the main text (not only the appendix) and explicitly mentions the anisotropic case as open.

18 [M4] P2 Concurrent Muon Literature Positioning

Due: April 27 **Cost:** $0 **Effort:** 2 hr

What to do

The related-work section (`v5_related_work-11.tex`) has a concurrent-Muon paragraph with partially unresolved citations. The two structural-distinction sentences (why classification and mirror-descent machineries do not transfer) are present in the introduction but must also appear in related work.

1. Search arXiv for concurrent Muon implicit-bias work (2025–2026): "Muon optimiser implicit bias regression," "spectral flattening polar map," "matrix regression implicit bias optimizer." Add any relevant papers to `references-3.bib`.
2. Confirm the following two sentences appear in the concurrent-work paragraph (if not, add them):
   - *Why classification doesn't transfer:* margin-boundary geometry vs. affine interpolation manifold.
   - *Why mirror descent doesn't apply:* the polar map is not the Riemannian gradient of any norm (Lemma A.1).
3. If no concurrent work is found, add: "At the time of submission, we are not aware of concurrent work settling the regression implicit bias of Muon."

19 [M5] P3 Discrete-Time Empirical Validation (Optional Campaign C4)

Due: May 3 (only if Fixes M1–M4 are complete) **Cost:** $100–$300

Background

Theorem A is stated for continuous-time Muon. The discussion (`v5_discussion-7.tex`) acknowledges the discrete-time gap. The B_{crit} formula and ATSR phase diagram provide indirect discrete-time evidence, but a direct tracking experiment would close the gap cleanly. This fix is **optional**: skip entirely if any of M1–M4 is unfinished on April 30.

Deliverables (if pursued)

1. Run discrete-time Muon on the same (d, r, N) configurations as Campaign C1, with $\eta \in \{0.1, 0.01, 0.001\}$, 3 seeds each. Compute $\max_t \|\sigma_i^{\text{discrete}}(t) - \sigma_i^{\text{ODE}}(t)\|$ and verify $\mathcal{O}(\eta)$ scaling. Save as `scripts/campaign_c4_discrete.py`.
2. Add Figure C4: tracking error vs. η on a log-log plot with $\mathcal{O}(\eta)$ reference line.
3. Add one sentence to Theorem A or its corollary: "Campaign C4 confirms discrete-time Muon tracks the ODE to $\mathcal{O}(\eta)$ across all tested configurations (Figure C4)."

20 [M6] P3 Anonymise Muon Paper (if submitted as supplementary)

Due: May 3 **Cost:** $0 **Effort:** 30 min

Deliverables

1. Anonymise the author block in `v5_main-9.tex`.
2. Run: `grep -rli "poggio\MIT`$\|CBMM\|Abdelmoneum\|Beneventano"v5_*.tex|andremoveallhits.$

2. Strip PDF metadata (title, author, creator fields).

Master 11-Day Schedule

Day	Date	Tasks	Fixes
1	Apr 24	Correct Frobenius error in tech report and verify Muon paper wording; run 4 baseline API calls for Table 3; parse DAG logs for Table 2; parse cost logs for §6.3; send expert verification package.	T1, T3, T4, T7, T11, M1
2	Apr 25	Fill Theorems A(iii) and B from Muon v5 into tech report; fill Proposition 3 numbers; begin bib rebuild (Muon entries first).	T2, T8
3	Apr 26	Complete bib rebuild; replace all `?` in both documents; extract Lyapunov council artefacts.	T6, T8, M2
4	Apr 27	Configure and launch 6 ablation partial runs; add ATSR results paragraph and Figure 3 to tech report; resolve concurrent Muon citations.	T5 (launch), T9, M4
5	Apr 28	Build Implementation Maturity Table E.1; write Theorem B anisotropic discussion or launch theory sub-run.	T10, M3
6	Apr 29	Collect ablation results; write Appendix F skeleton; write Appendix G (Lyapunov council).	T5 (write-up), T6 (finalise)
7	Apr 30	Write §6.1–§6.3 with all filled numbers; complete Appendix F with available scores; zero `[GAP]`/`[NUM]` in tech report.	T3, T4, T7 (integration)
8	May 1	Full tech-report prose rewrite of §5–§6; update abstract draft and contribution list; optional: launch Campaign C4.	T14, M5 (optional)
9	May 2	Apply external verifier response; complete NeurIPS checklist; expand Broader Impact.	T11, T13
10	May 3	Full anonymisation of both documents; final abstract polish ($\leq$ 250 words); compile clean PDFs; run all grep checks.	T12, T14, M6
11	May 4	**Submit abstract by 23:59 AoE.** Continue polishing for full paper deadline (May 6 AoE).	—

Cut list if behind on April 30. Drop in order: (1) Fix M5 (Campaign C4), (2) Fix M3 Option A (theory sub-run), (3) Figure 2 Panel B (cost burn-down), (4) Table E.1 extra columns.

Never cut: T1 (wrong theorem is fatal) — T3 (Table 3 is the primary counterfactual) — T5 skeleton (at least Config A runs must complete) — T6 (Lyapunov artefacts are the council's key evidence) — T12 (anonymisation failure = disqualification).

Quick-reference completion check. Run before submission:

```
grep -cn "\[GAP\|\[NUM\|\[VERIFY" pAI_tech_report.tex   # must be 0
grep -c "(?)"                     pAI_tech_report.tex   # must be 0
grep -rn "(?)"                    v5_*.tex              # must be 0
grep -i  "poggio\|MIT\|CBMM\|Abdelmoneum\|Beneventano\|Perseus" \
         pAI_tech_report.tex                            # must be 0
```